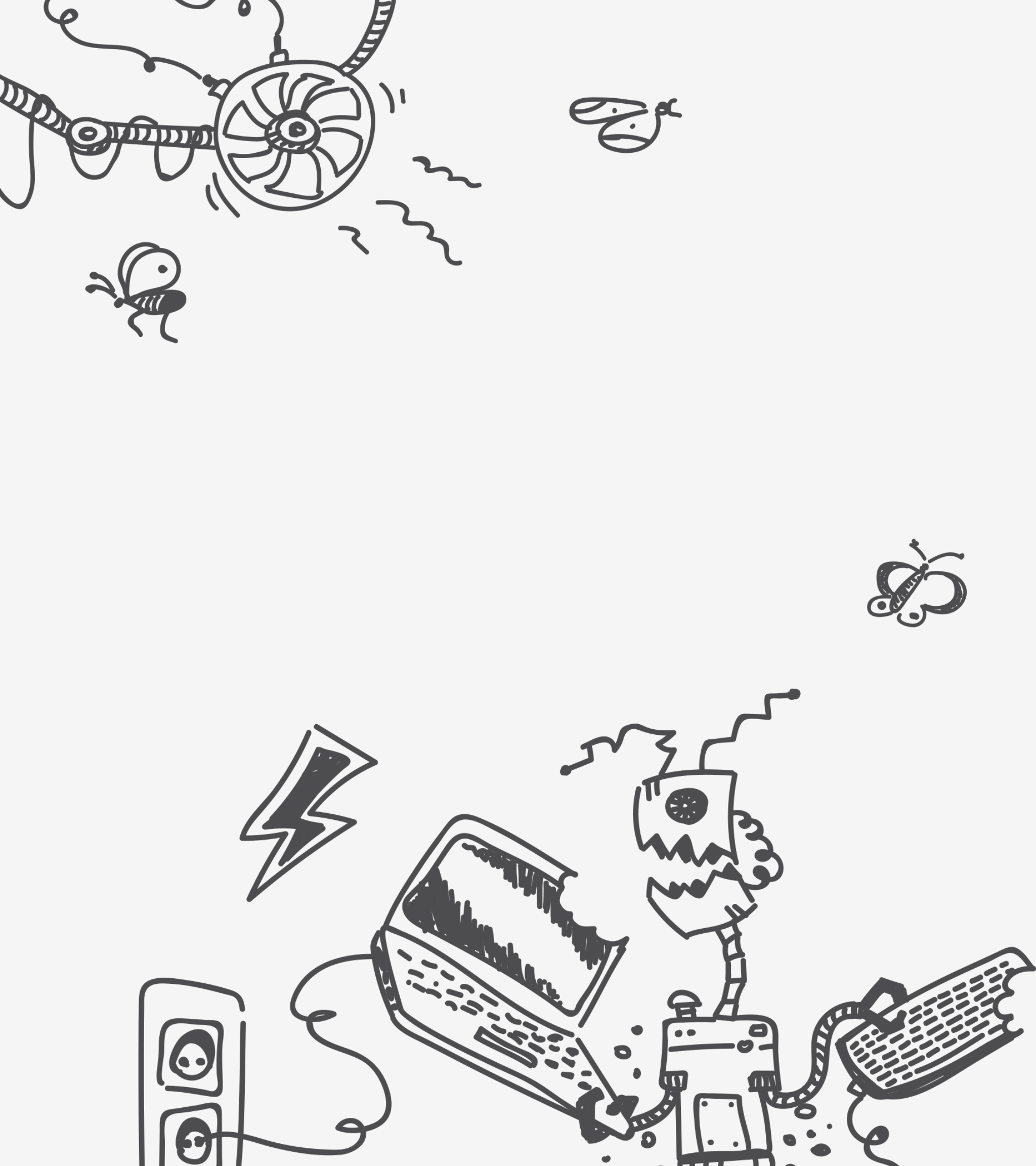
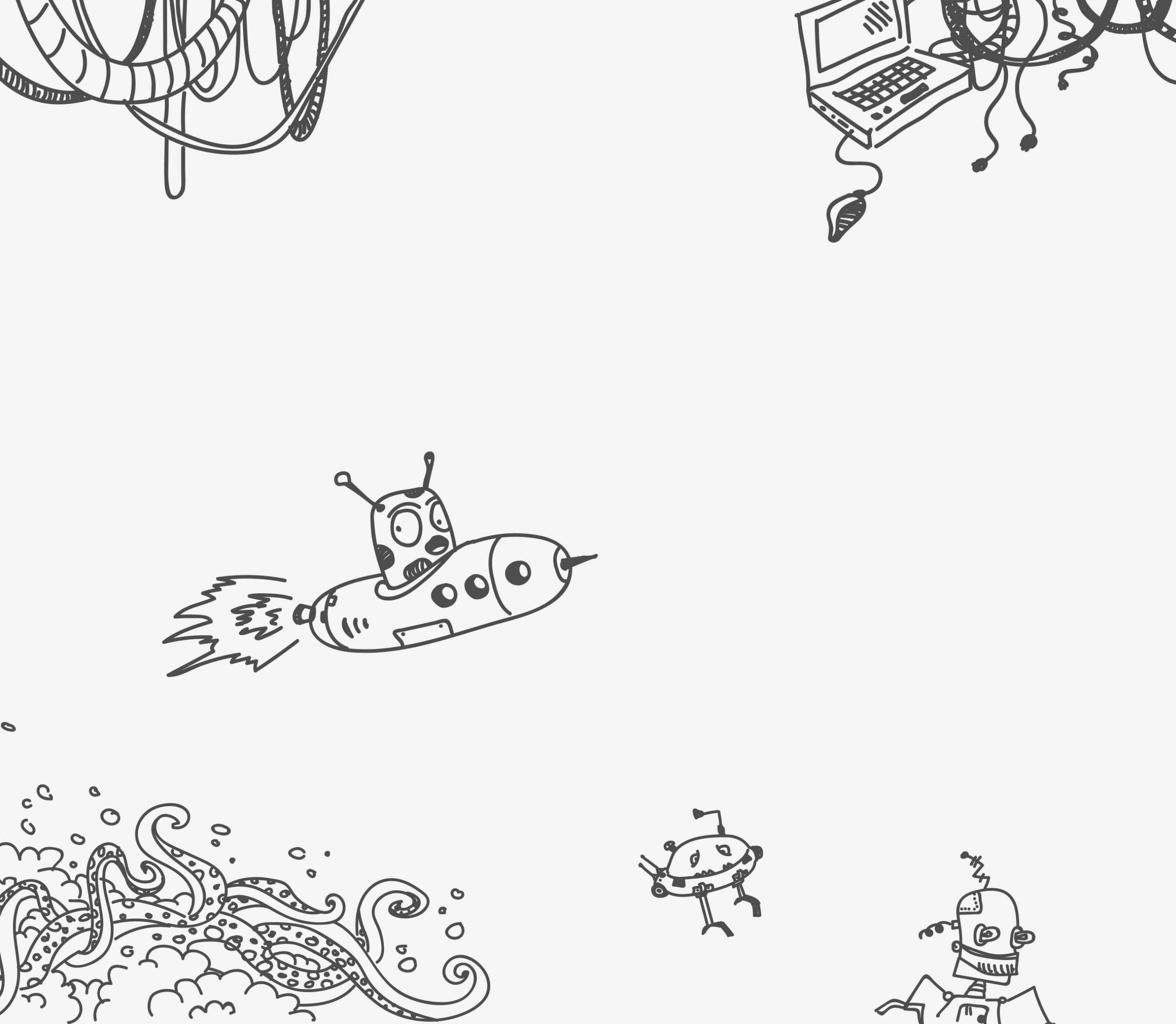




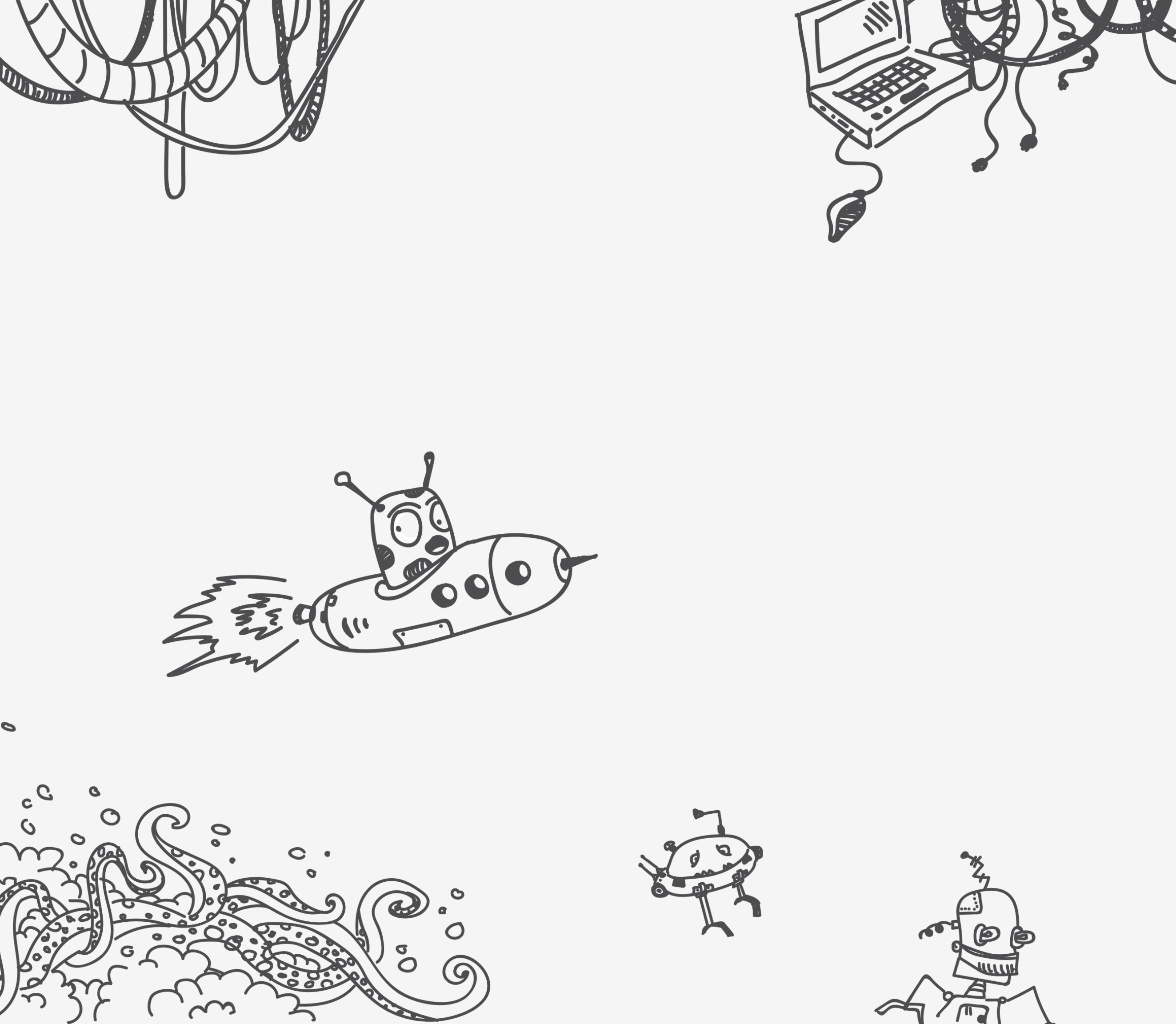
Linear regression



Introduction

Introduction

Let's start our adventure with machine learning in earnest. In this presentation, we will learn about one of the simplest machine learning algorithms. Nevertheless, the concept of linear regression does not only appear in **machine learning** topics. Those who were dealing with statistics before the *Data Analyst* course have certainly heard of this forecasting tool.



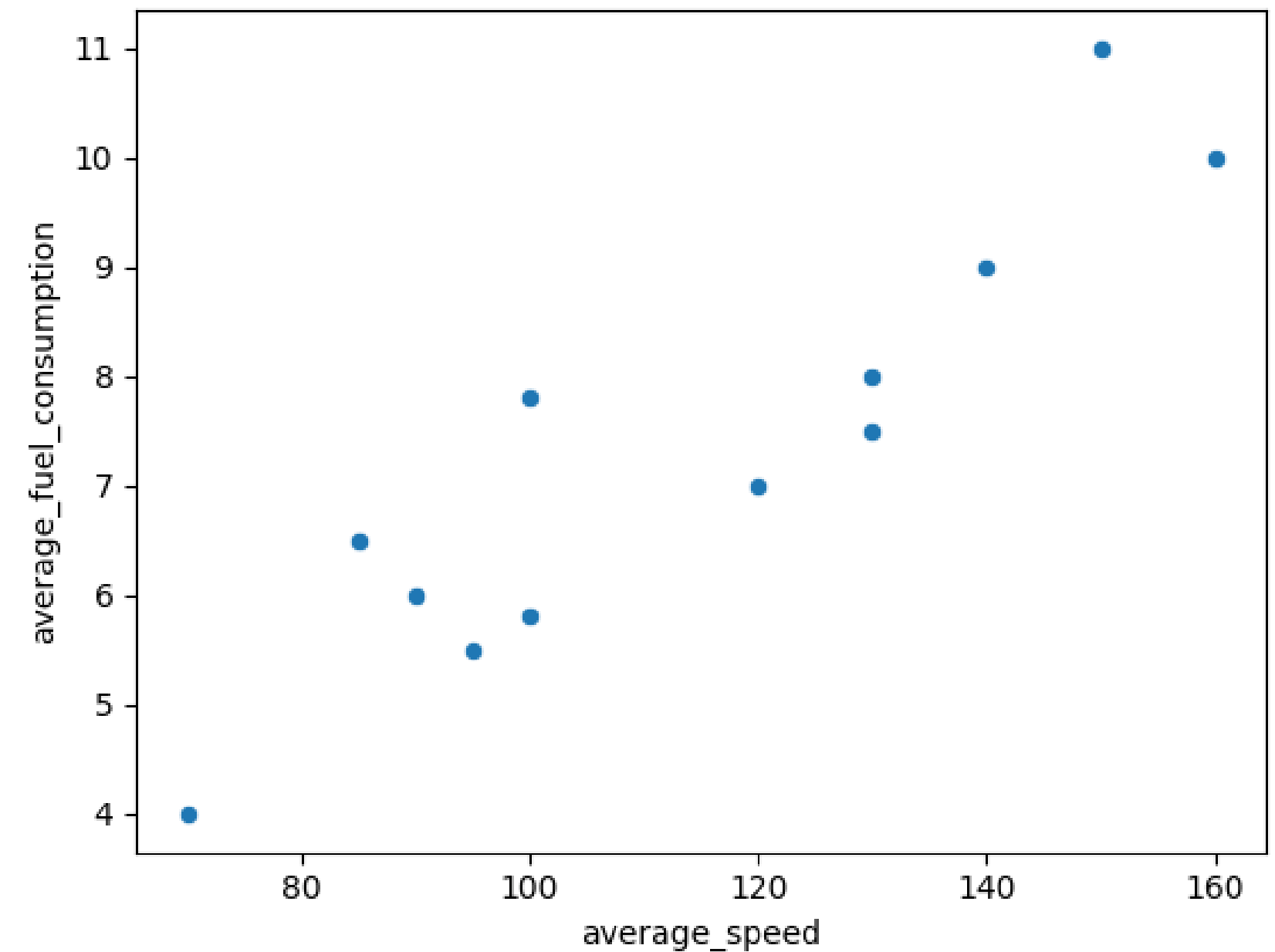
Linear regression - what is it?

Linear regression - what is it?

Linear regression is an algorithm whose task is to fit a straight line to the data. Speaking of a straight line, means that the algorithm assumes that the relationship between the independent variables (i.e. our characteristics) and the dependent variable (i.e. what we want to predict/estimate) is linear. Of course, this relationship will never be perfectly linear, but it will come close to such.

The best way to understand how linear regression works would be to analyze a simple example based on a real-world problem.

Suppose we have data on the routes traveled by one car. We have information about the average speed on a given route. The task is to estimate the average fuel consumption of the car for every 100 kilometers driven. For some routes we



Linear regression - what is it?

Our task is to fit a straight line to the points in such a way that it best describes the relationship between average speed and average fuel consumption. We will try to find the corresponding values of the coefficients a and b in the linear function:

$$y = ax + b + \epsilon$$

Where:

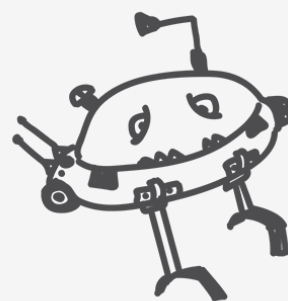
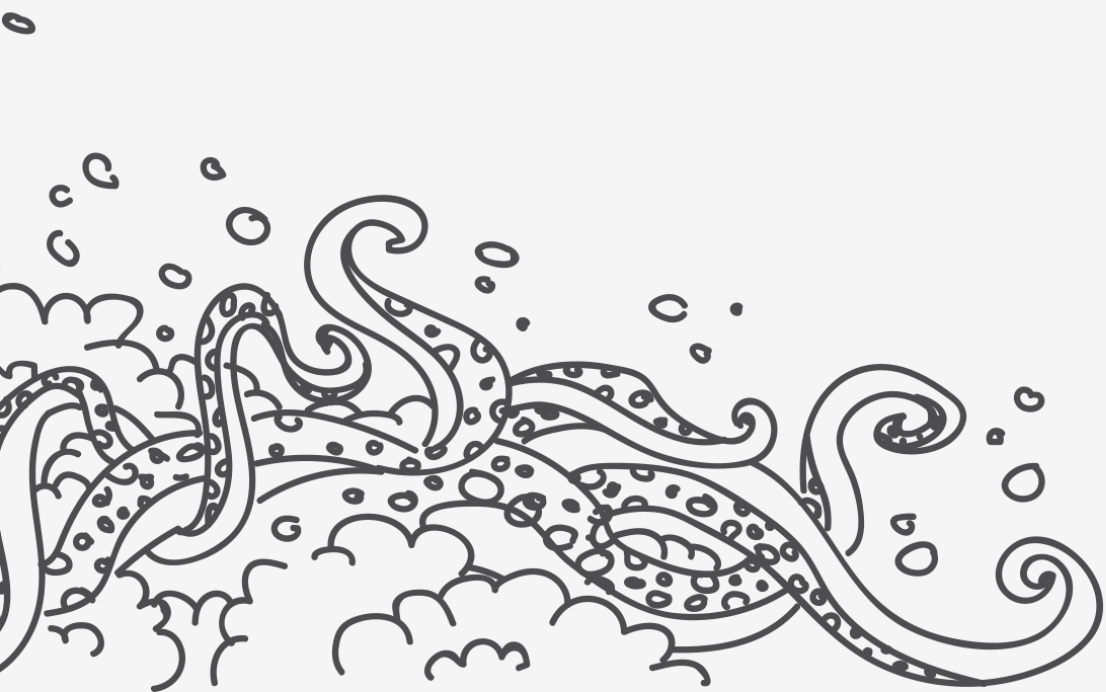
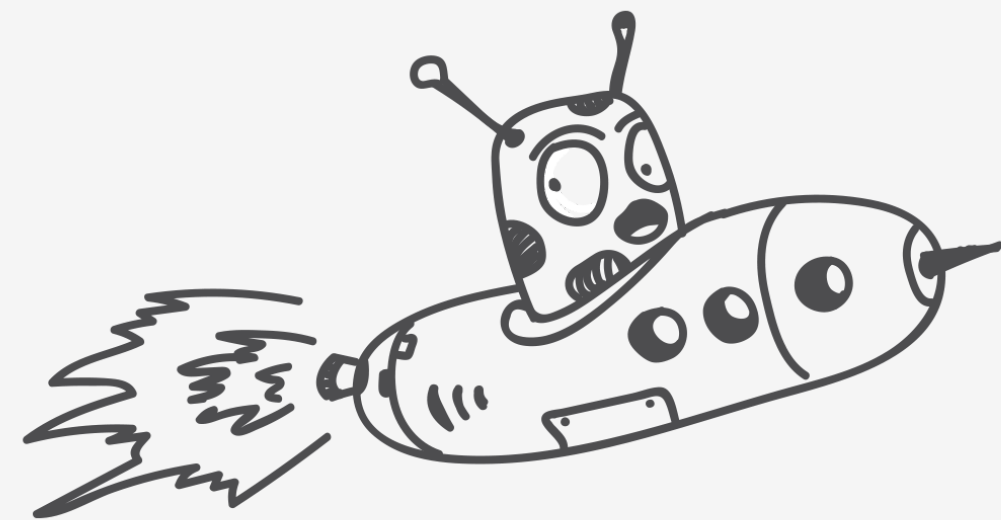
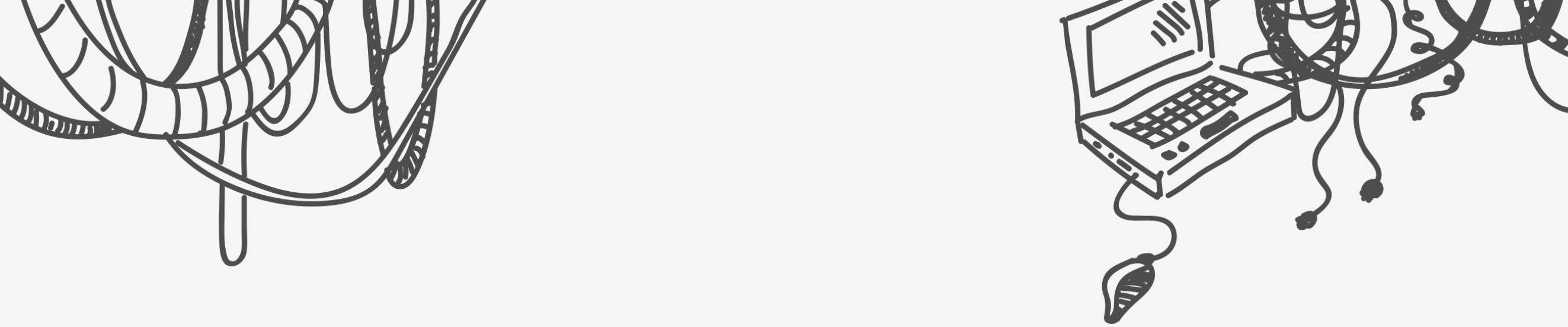
y - dependent variable (average fuel consumption on a given route),

x - independent variable (average speed of the car on a given route),

a - *slope coefficient*,

b - *intercept*

ϵ - random error.

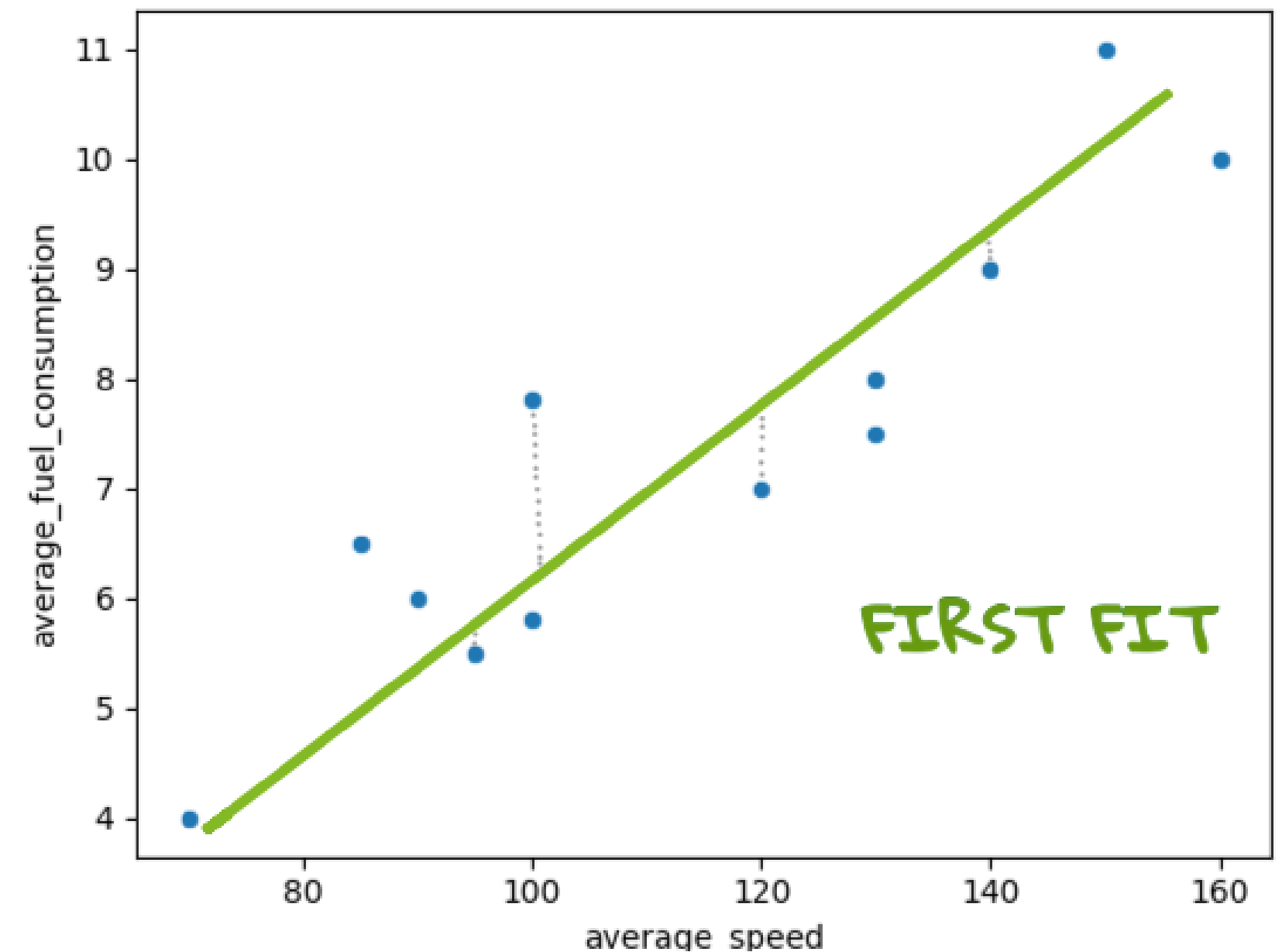


**How do we find the
best straight line fit to
the data? Least
squares method**

Least squares method

Theoretically, looking at a scatter plot showing the relationship between our variables, we could try to draw various lines that would somehow fit our data.

Our goal, however, is to find the optimal straight line. To achieve this goal, we will use the method of least squares. This method involves finding the minimum for the sum of squares of the differences between the observed values and the values from our equation. Let's try to understand it by looking at the graphic.



Least squares method

In the graphic, the blue points are our observations, and the green line is the straight line we are looking for. What we care about is finding such a green line, at which the sum of the squares of the distances of the points from the straight line is the smallest.

So how do we determine the coefficients in our equation?

For one variable, which is the case we are considering, the determination is quite simple. We need to find the minimum for the sum of squares:

$$\min \sum_{i=0}^n = (Y_i - y_i)^2$$

Where:

Y_i - the actual value,

y_i - the value predicted by the model.

Least squares method

We can find the minimum by using the operation of differentiation, thus calculating the derivatives of the function after **a** and **b** (that is, the parameters we are looking for). In mathematics, derivatives are a way of measuring how quickly the value of a function changes as we increase the value of the argument. We can think of it as the slope of a tangent line to a function's curve - the steeper the line, the faster the function's value changes. Every function - quadratic, root, linear, etc. - has its own derivative formula.

However, we will not go into the determination of derivatives at this point. We mention it to foster your awareness of where the following formulas come from. During your daily work with algorithms, you will not need the knowledge of the exact calculations. Below you will find the formulas for the directional coefficient of regression and the intercept:

$$a = \frac{\sum (x_i - x)(y_i - y)}{\sum (x_i - x)^2}$$

$$b = y - ax$$

Least squares method

What do the above coefficients mean? The directional coefficient of regression (**a**) tells you what is the effect of a unit change in **x** on the variable **y**. If **a** has a positive value, then an increase in **x** by a unit means that we can expect **y** to increase by **a** units. If **a** has a negative value, then as **x** increases by 1, **y** decreases by roughly **a** units. Why roughly? Linear regression will never predict values with 100% accuracy. What it results in is an approximation of the **y** variable, subject to error.

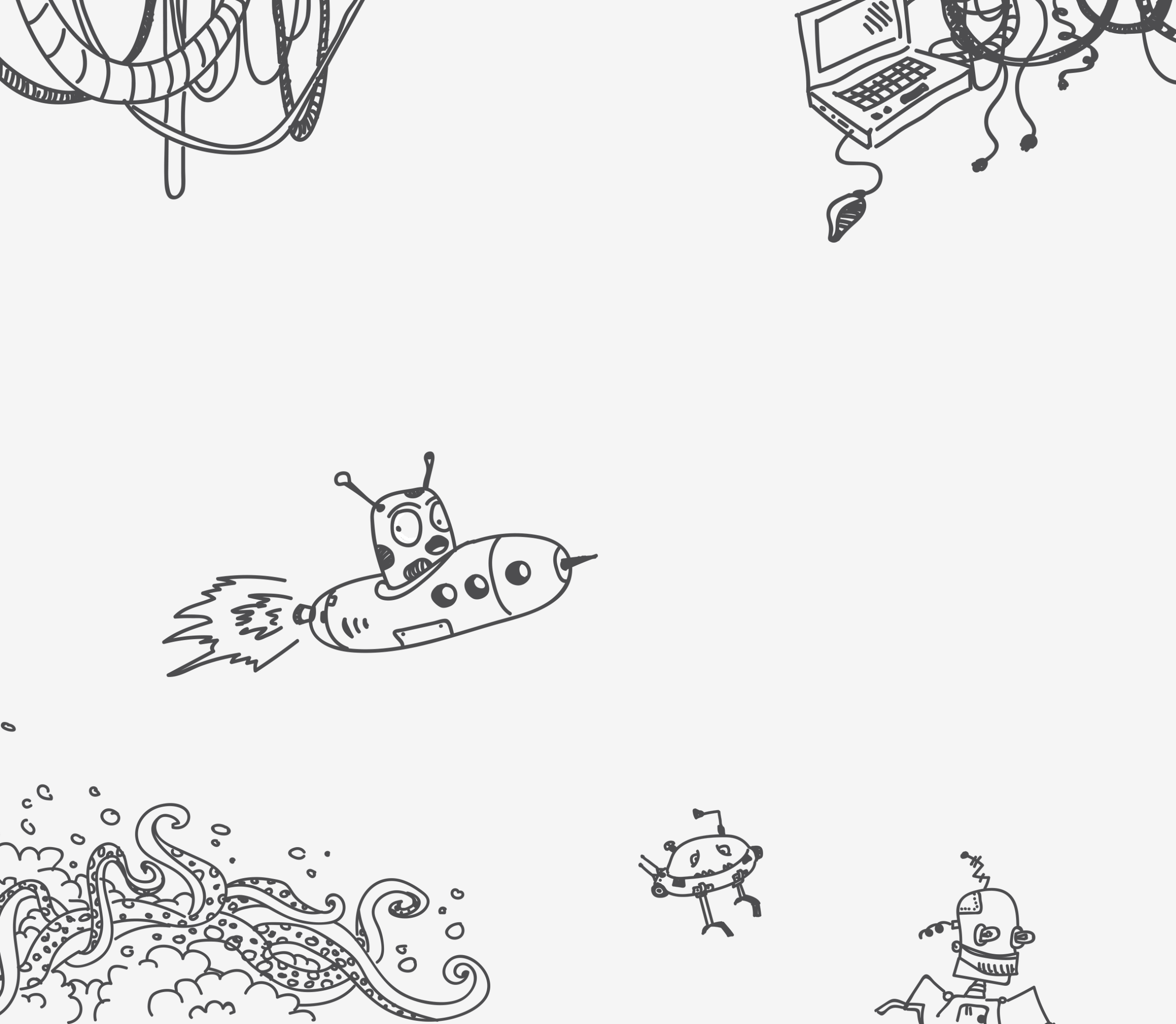
The intercept (**b**) tells us what value of **y** we should expect for **x** being 0. Let's analyze the regression formula we obtained for our problem:

$$y = 0,06428027 * x + 0,00300286$$

Least squares method

Let's say our speed was 120km/h, then we can estimate the average consumption of the car per 100km by substituting speed as x into the above formula, which gives us 7.716l / 100km.

Assuming that our speed (x) was 0 km/h, that is, the car was not moving, then the average consumption is 0.003l / 100 km. Theoretically, a stopped car should have an average combustion of 0 liters per 100 km. In this case, the value of the intercept has some overlap with reality. Suppose, for example, we would like to determine the price of an apartment based on its area. We determined the linear regression formula $y = 10000 * x + 3000$. This means that an apartment with an area of 0 m² costs 3000 PLN. In this case, the information does not make sense, because an apartment of 0 m² cannot exist.

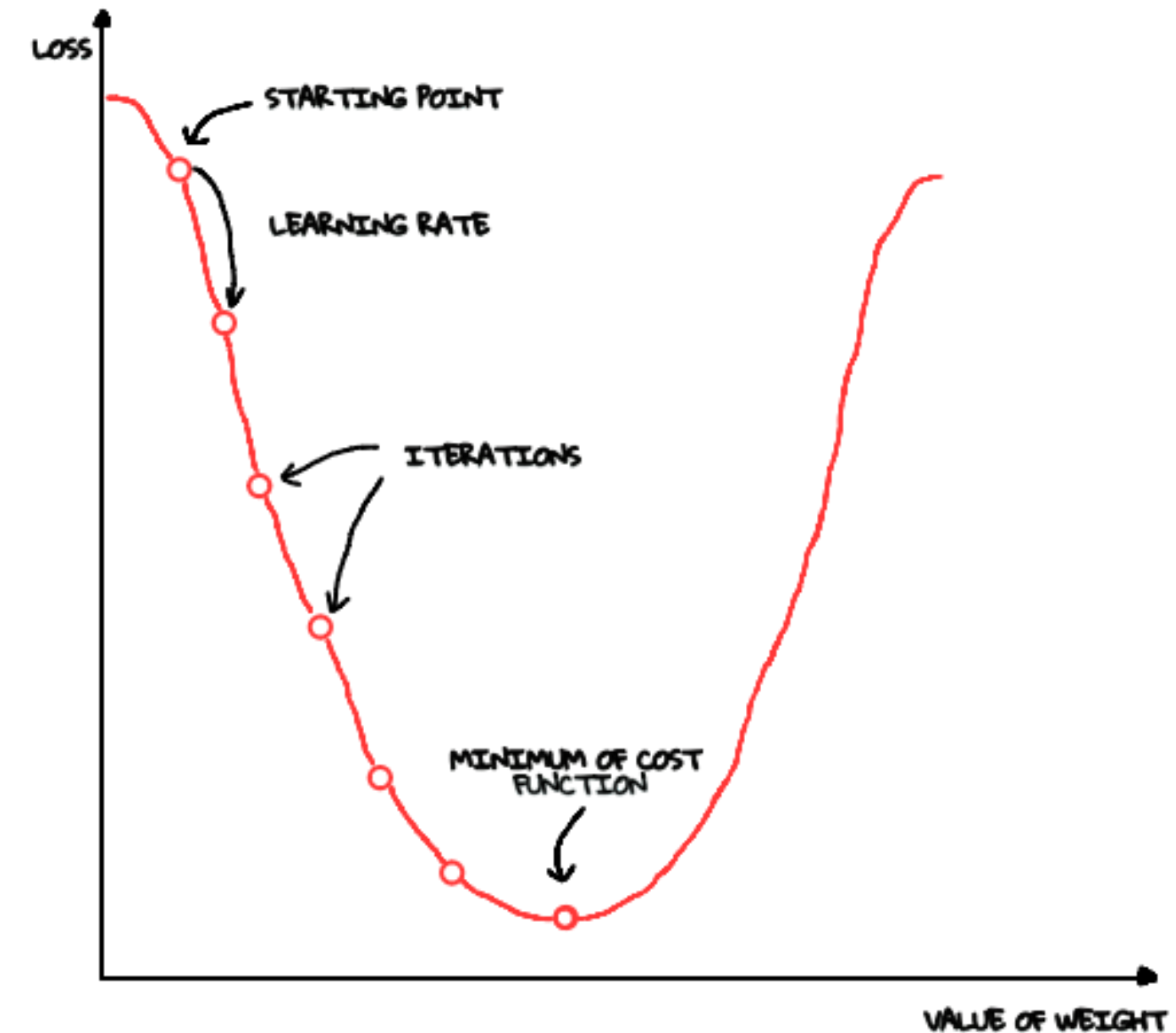


Least squares method vs. simple gradient

13

Least squares method vs. simple gradient

In our example, we used the method of least squares to minimize the sum of squares of the differences between the actual values and the values predicted by the model. Nevertheless, the cost function will not always be the sum of squares of the differences. In this case, we must use another solution to minimize this function. Simple gradient can help here: it is a more general and flexible method for minimizing the cost function. The simple gradient is an iterative method, that is, one where we repeat the same operation for an assumed number of times until we reach a solution of a certain accuracy or meet a certain condition.



Least squares method vs. simple gradient

The simple gradient is calculated by determining the partial derivatives of the cost function with respect to each model parameter (in our case, **a** and **b**). Based on the derivatives of the functions, the directions in which it is worth changing the values of the model parameter to minimize the cost function are calculated. To do this, we calculate the simple gradient for the assumed initial values of the parameters **a** and **b**. Let's assume that **a** and **b** are 0 at the beginning. We also assume that with each iteration our parameters will change 0.00001, and the number of iterations will be 10000. According to our assumptions, we calculate the partial derivatives and update the values of the coefficients **a** and **b**. To do this, we will use code written in Python.

Least squares method vs. simple gradient

```
X = fuel_data["average_speed"]
y = fuel_data["average_fuel_consumption"]

# Define starting values for "a" and "b"
a = 0
b = 0

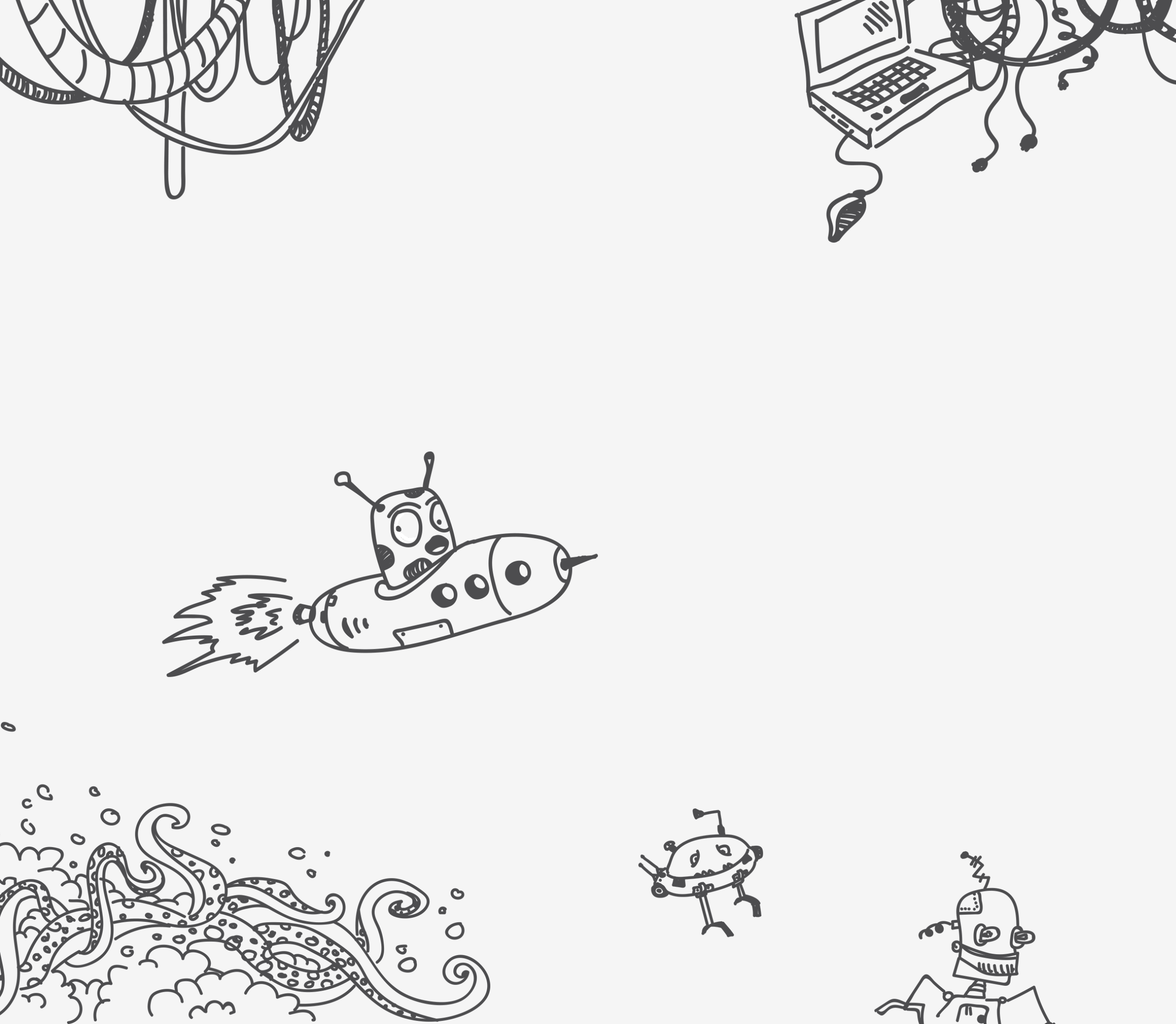
# Define learning rate and number of iterations (epochs)
L = 0.00001
epochs = 10000

# Number of elements in X
n = float(len(X))

# Performing Gradient Descent
for i in range(epochs):

    y_pred = a * X + b

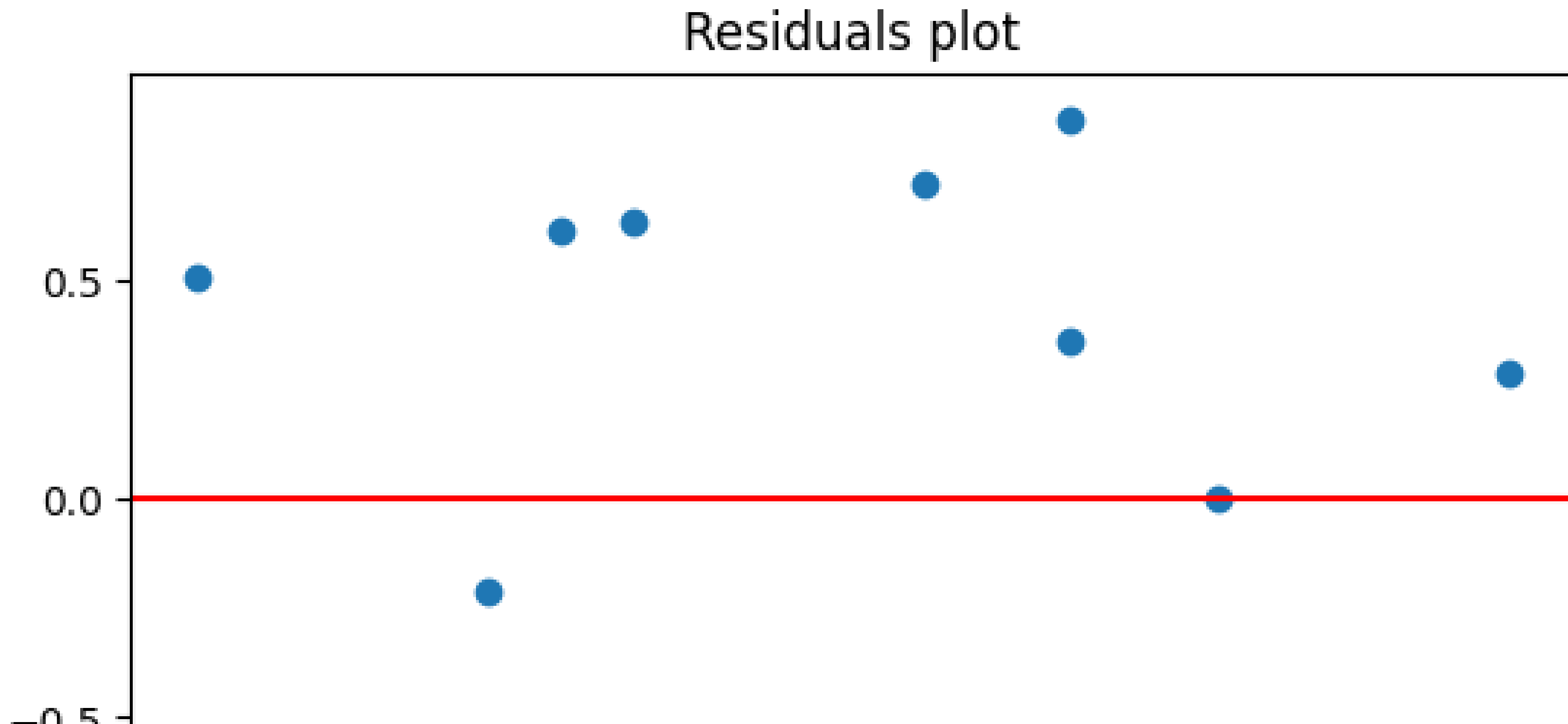
    # Derivative of function with respect to a
    derivative_a = (-2/n) * sum(X * (y - y_pred))
```



**Does our linear
regression model
make sense?**

Does our linear regression model make sense?

Having the equation determined, as well as the actual values, we can determine what the differences are between the values calculated from the model equation and the measured (actual) values. Assuming that the line determining the model is our reference point, then we can plot the actual values around it to visualize the model error:



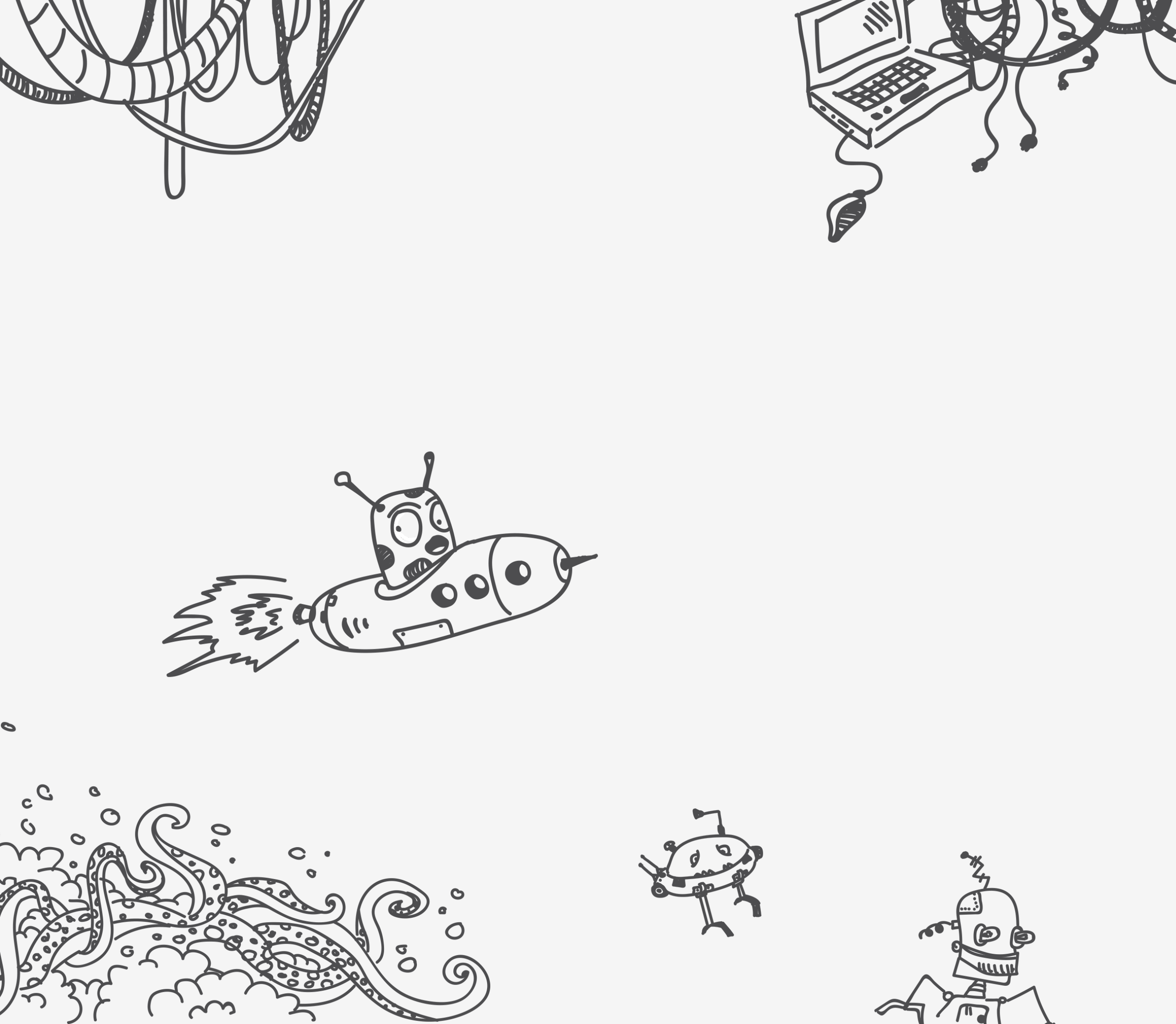
Does our linear regression model make sense?

The greater the scatter of these values, the less chance that the linear regression model is a good choice for solving our problem. There are two ways out of this situation:

- making sure we have a larger number of independent variables that will allow us to better determine the dependent variable,
- changing the model to another one better suited to the problem.

In addition to analyzing the residuals graph, we can determine the coefficient of determination R^2 , which takes values from 0 to 1. When R^2 is 1, it means that the model is perfect. We will explain where this value comes from in more detail during the presentation on regression model metrics.

For our model, it was **0.83**, which we can interpret as the situation that the speed of our car explains 83% of why the car burns a given amount of fuel per 100 kilometers on average.



Linear regression vs. outliers

Linear regression vs. outliers

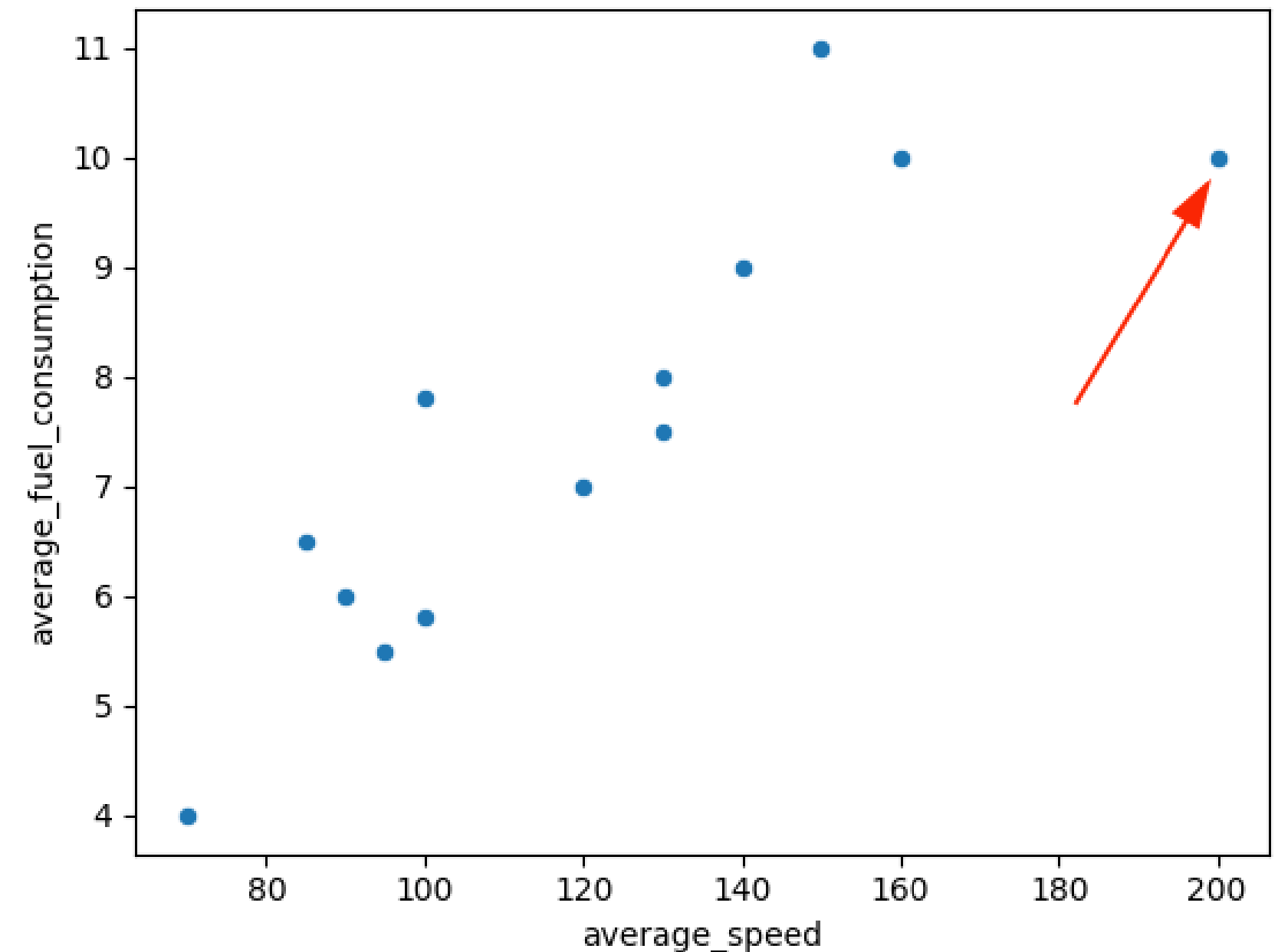
Linear regression is susceptible to outliers. Outliers are observations that deviate significantly from the rest of the data in the dataset. They can have a significant impact on the outcome of a linear regression, since linear regression relies on minimizing the sum of squares of the differences between the values predicted by the model and the actual values.

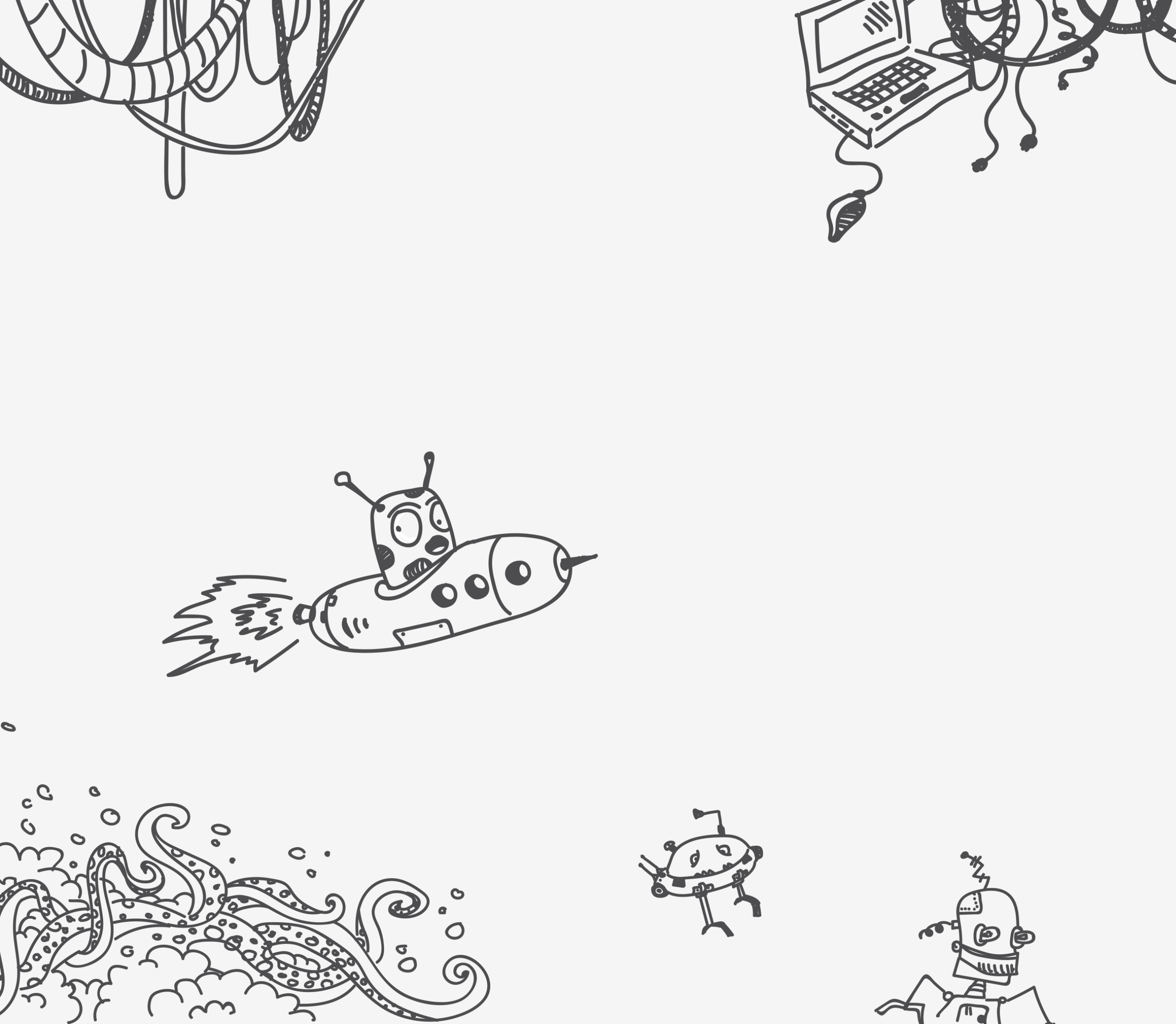
If there are outliers in the data set, they can significantly affect the linear regression model because the model will try to fit a line that best fits all the data, including the outliers. As a result, the model may predict values that are far outside the range of actual values, leading to errors in predictions.

Linear regression vs. outliers

Let's see how our model will be affected by training it with data to which we add an outlier. Let's add data on a route during which the consumption was 10l and the speed was 200km/h.

Training our model on the updated data, gives an R^2 coefficient of determination equal **0.76**. Based on this we can see that the quality of our prediction has decreased by adding an outlier to the set. Therefore, it is important to remember that if you want the best possible linear regression model, you should avoid training the algorithm with data containing outliers.





**Can linear regression
use only one factor to
predict values?**

Can linear regression use only one factor to predict values?

Of course **not**. We started with a simple example, so as to best understand how a linear regression model works. The problems we analyze will practically always consist of more extensive data, containing many variables that affect the predicted outcome. Then the describing relationship is more complex, as is the determination of the directional coefficients of the regression:

$$y = a_1x_1 + a_2x_2 + a_3x_3 + \dots + b + \epsilon$$

The number of coefficients **a** in the above equation depends on the number of variables in our data set.

So the number of variables can be arbitrary, but it is important to remember that such a situation is not always good and valuable. In the case where there is a high correlation between two independent variables, we should drop one of them. Using two variables at the same time does not bring any gain in terms of accuracy of the linear regression model. We have already described this phenomenon during the prework, and we call it *multicollinearity*.

Summary

During the presentation, we talked about what linear regression is and what problems we can solve using it. We know what formula describes linear regression, and how the parameters of linear regression, namely the directional coefficient and the intercept, are determined. We learned about the method of least squares and simple gradient as a way to find the optimal value of the cost function. We introduced the concept of coefficient of determination as a measure of the quality of a regression model. With theoretical knowledge, we can move on to the practical part - we will build a linear regression model in Python.